

Stage 3 Judges' Rubric - DECI Levels 4 & 5

Combined Advanced Project - Stage 3 Judge Evaluation

Team Name / #	Project Name	Judge Name	Date

How to Score Every Row Enter ONE number from 1 to 5. Every row is worth 5 points; harder sections have more rows and therefore more weight.

1	Weak	Missing, incorrect, non-functional, or little/no meaningful evidence.
2	Developing	Partially demonstrated; important gaps remain or substantial support is needed.
3	Meets Expectations	Core requirement is present, understandable, and works at an acceptable/basic level.
4	Strong	Complete, accurate, well integrated, and supported by good evidence; only minor issues.
5	Excellent	Highly effective, reliable, purposeful, well-tested/polished, and clearly justified by the team.

Judge the final Stage 3 work, using the previous-phase requirements as the baseline. **Flow:** reward evidence, correct integration, and purposeful design - not complexity for its own sake. **Testing:** look for real trials/results, not only claims. **Hardware:** reward the best-fit use of components and reliable integration. **Teamwork & presentation:** look for shared ownership, clear communication, and the ability of multiple members to answer questions.

Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Problem Analysis & Complete Solution			Section Weight: 10 points
Problem, research & opportunity	Defines the real problem, target users, existing alternatives, supporting research, and the specific value/opportunity for the solution.		
Complete end-to-end solution	User journey and system behavior form a complete flow from data capture/input through processing, user action, and feedback/logging where appropriate.		

Architecture, Hardware & Embedded IoT			Section Weight: 25 points
System architecture & data flow	Hardware, ESP32/ESP32 Camera, communication, AI/software processing, and user interface are connected in a technically coherent architecture.		
Purposeful hardware selection	Physical platform, sensors/actuators, ESP32, and ESP32 Camera roles are justified by the problem; components are not added without purpose.		

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Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Architecture, Hardware & Embedded IoT (continued)			Section Weight: 25 points
Circuit & embedded logic	Circuit, input processing, states/events, actuator responses, and control logic are technically sound and match the implementation.		
IoT / communication integration	Data transmission, cloud/local processing, and device-to-system communication are reliable and appropriate for the use case.		
Best hardware use, reliability & scalability	Hardware capabilities are used efficiently; system performs consistently and design choices consider reliability and practical scaling.		
AI, Data Processing & Data Science			Section Weight: 20 points
Data pipeline	Data collection, preparation, processing, and decision flow are clear and technically appropriate.		
AI/ML approach & performance evidence	Chosen AI/ML method fits the task; team shows relevant outputs, confidence/accuracy evidence, or justified evaluation.		
Data science insight	Collected project data is organized/visualized to produce a useful insight that supports a real user or business decision.		
Integration of AI/data into system behavior	AI/data results trigger meaningful dashboard updates, alerts, predictions, recommendations, or other actions in the same solution.		
Product, UX, Web & Cybersecurity			Section Weight: 15 points
Product identity & professional visualization	Brand/UI is consistent and polished; mixed Level 4-5 team includes strong product visuals/digital assets appropriate to Level 5 depth.		

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Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Product, UX, Web & Cybersecurity (continued)			Section Weight: 15 points
Web application integration	Web/app interface exposes the correct live/history/control information from the hardware/AI system and supports the intended user flow.		
Cybersecurity design	Authentication/authorization, access control, communication/data protection, and relevant risks are addressed in a way that fits the architecture.		
Business, Testing & Improvement			Section Weight: 15 points
Business model & product planning	Customer, value proposition, pricing, competitors/alternatives, and delivery approach are complete and realistic.		
Testing evidence & measurable success criteria	Team tests under relevant conditions and shows evidence such as logs, accuracy results, repeated runs, failure cases, or comparison to ground truth.		
Iteration, reliability & improvement	Test results lead to concrete improvements; team can explain limitations, reliability concerns, and next technical/product steps.		
Teamwork & Presentation			Section Weight: 15 points
Cross-discipline teamwork & ownership	Hardware, software, AI/data, design, security, and business work are integrated; members have visible contributions and shared ownership.		
Technical communication & decision justification	Team explains architecture and tradeoffs clearly and can justify why technologies/components were chosen.		
Presentation, live demo & QA;	Presentation is polished and concise, demo evidence is convincing, and multiple members answer judge questions confidently.		

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Specific Judging Task	What the Judge Should Look For	Score (1-5)	Judge Note
Extra Requirements / Optional Bonus			Optional Bonus - max 15
Advanced AI/ML feature	Optional advanced AI feature goes beyond the core AI/data pipeline and is supported by evidence or a credible prototype.		
Advanced IoT / security feature	Optional advanced cloud, real-time communication, automation, monitoring, or security capability strengthens the same architecture.		
Level 5 advanced security / reliability	Optional Level-5-depth feature such as stronger authentication, secure protocols, monitoring, risk management, or reliability safeguards is demonstrated with evidence.		

Score Summary

Section	Maximum	Section Subtotal	Notes
Problem Analysis & Complete Solution	10		
Architecture, Hardware & Embedded IoT	25		
AI, Data Processing & Data Science	20		
Product, UX, Web & Cybersecurity	15		
Business, Testing & Improvement	15		
Teamwork & Presentation	15		
CORE TOTAL	100		

OPTIONAL BONUS (record separately, max 15)	
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